

AI-Assisted Edge Vision for Violence Detection in IoT-Based Industrial Surveillance Networks

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Abstract—Analyzing surveillance videos is mandatory for the public and industrial security. Overwhelming growth in computer vision fields has been made to automate the surveillance system in terms of human activity recognition, such as behavior analysis and violence detection (VD). However, it is challenging to detect and analyze the violent scenes intelligently to fulfill the notion of Industrial Internet of Things (IIoT)-based surveillance buoyed by constrained resources to reduce computational power. To tackle this challenge, in this article, an artificial intelligence enabled IIoT-based framework with VD-Network (VD-Net) is proposed. First, the input video frames are passed to lightweight convolutional neural network model for important information collection including humans or suspicious objects such as knives/guns. Upon suspicious object detection, an alert is generated as an earlier VD in IIoT network while the information is shared with concern departments. Only the frames with objects are forwarded to cloud for detail investigation where features are extracted using convolutional long short-term memory (ConvLSTM). The latter from ConvLSTM is propagated to gated recurrent unit for final VD. The conducted experiments and ablation study on the existing surveillance and nonsurveillance datasets empirically validate the effectiveness of the proposed VD-Net by improving 3.9% increase in the accuracy compared with the state-of-the-art VD methods.

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I. INTRODUCTION

OVER THE past few years, extremely congested areas, such as public streets, subways, industries, and banks are widely monitored via surveillance systems to ensure public safety in smart city [1]. However, their effectiveness and efficiency are questionable since they require humans' involvement and their skills in efficient management and monitoring.

A report published in [2] states that the interpersonal violence has killed 48% of the people in South Korea by 2015 where 25% deaths have been occurred due to the use of sharp objects, such as knives, etc. Similarly, about 1.2 million violent activities have been happened in the United State by 2019 that has encompassed fighting, aggressive actions, and mass shooting. Hence, there is a significant increase in the deployment of surveillance cameras in both public and private sectors that has made their automatic monitoring impossible. So far, the detection of certain activity as normal or abnormal, is quite difficult to handle intelligently. Thus, an artificial intelligence enabled industrial Internet of Thing (IIoT) network-based violence detection (VD) framework will further enhance the public and industrial security and safety level in an automated fashion through eliminating the manual surveillance systems. Such systems will help the authorities related to law-enforcement by alarming them against the detected violence. Similarly, VD in videos is an imperative step towards the deployment of a system that will be capable of distinguishing the normal behavior from an abnormal/violent, e.g., normal behaviors can be categorized as routine behaviors including jogging, walking, and hand shaking while violence is an activity that hurts or applies physical force to affect a human or damage the state of certain object, i.e., person to person or person to many as fighting. Such type of activities needs to be detected in the early stages to prevent crimes and aggressive actions in an automatic and intelligent manner.

For this purpose, researchers with diverse background are now investigating new ways to thoroughly analyze the activities and automate their recognition in surveillance videos [3]. This analysis has been done in terms of activity recognition [4], video summarization [5], and reidentification of persons to recognize

individual's activity in video surveillance [6]. However, a little attention has been paid towards the VD in industrial surveillance and has been neglected in IIoT network. In the similar manner, the researchers have applied traditional handcrafted features or convolutional neural network for VD in surveillance and nonsurveillance videos, but it remained a big challenge to fulfill the notion of IIoT network in the detection of violent scenes. Despite of recent advancements in VD, there are no effective solutions to execute these VD methods in video surveillance. Thus, an IIoT-based network that is connected to vision sensors which screens the frames to collect important information, such as humans or suspicious object, will further assist the law enforcement services to prevent the crime in smart city. In such system, upon suspicious objects detection, the information is shared in the IIoT Network to analyze the object and follow the safety concerns. Implementing this setup will help in monitoring the activities at multiple location cameras in an automatic way to handle the abnormal events.

Various handcrafted and deep learning-based techniques, algorithms, and datasets for VD in surveillance and nonsurveillance environment are surveyed in [7]. The surveyed methods have distinct approaches which are comprised of different parameters, and features, such as Hough forest (HF) [8] and Lagrangian function [9]. Although, there exists several challenges and problems in the SOTA techniques including high computation, viewpoint, illumination, dealing complex crowd scenes, and intensity variations. Similarly, majority of these methods are based on nonsurveillance approaches and have no inclusion of IIoT network to assist VD analysis. Considering these facts, a method that ensures the inclusion of edge computing feasible with cloud server to analyze certain activity for violent scenes in industrial environment, will be more worthy. To assure this, we propose an AI enabled IIoT-based computationally intelligent framework with following core contributions.

- 1) Wires failure is one of the drawbacks in the traditional monitoring systems along with its installation. This setup causes huge processing and late response making it a tedious job for the authorities. To tackle this issue, we implement an AI enabled IIoT-based VD framework where the devices are connected to the Internet router that interact each other and share the information. Next, a cloud system is developed that assists the VD process for detailed investigation of violent scenes in industrial settings where the processing speed is 250 fps.
- 2) Processing large amount of video data generated from industrial surveillance needs an intelligent mechanism to acquire the relevant and actionable information for VD analysis on the edge. Excessive usage of bandwidth and energy have made SOTA VD methods comparatively costly and time consuming. Considering IIoT into account, we investigate a light-weight CNN model that is deployed over the embedded device to screen the frames for important information such as humans or suspicious objects. In case of suspicious object, this information is shared over the network and instead of all the frames, only

the important frames having objects are sent into cloud for detailed investigation. This approach makes the process smoother and comparatively intelligent.

- 3) Conventional VD approaches widely use handcrafted features or apply clustering and features descriptors, etc., that are not suitable for VD in long-term dependencies in the IIoT network resulting in lower accuracy. To overcome this issue, we employ ConvLSTM layers for VD in IIoT network for the first time that has the capability to encode the spatio-temporal co-relation from the industrial activity sequence to analyze the local motion between the frames. This step is completely supervised by the cloud server where the obtained features vector is propagated to a recurrent unit for learning the sequences.
- 4) To the best of our knowledge, this article employs gated recurrent unit (GRU) for the first time for learning the violent activity sequences in IIoT network by decoding the latter from ConvLSTM and make final decision for violent and nonviolent scenes in industrial setup. The empirical analysis of the proposed VD-Net over the publicly available datasets not only outperforms over SOTA but can be also deployed in IIoT-based indoor and outdoor surveillance.
- 5) Due to unavailability of well annotated real-world VD datasets, we introduce a new dataset comprising of 300 videos related to real-world industrial surveillance having violent and nonviolent classes. We believe, this dataset will play a vital role in VD domain.

The rest of this article is organized as follows. Section II covers the related literature while Section III discusses the proposed IIoT-based VD-Net. In Section IV, we cover the experimental results and their analysis. Finally, Section V concludes this article.

II. RELATED LITERATURE

This section covers the most recent work carried out in VD arena comprise of both traditional and deep learning-based methods that have obtained overwhelming attentions of the researchers.

A. VD Practice Using the Traditional Methods

Traditional methods include the practices of using handcrafted or low-level features. For instance, the method in [10] used appearance and dynamic as low-level features and extracted feature through local histogram of oriented gradient descriptor from RGB frames where motion regions obtained through optical flow were segmented. They coded these features into bag of words (BoWs) to remove the redundant information and finally detected the violent behavior. Similarly, Senst *et al.* [9] used Lagrangian fields of directions based on the appearance, background motion compensation, and long-term motion details to detect violence in crowded scenes. Using late fusion, they used BoW and proved that the temporal scale is crucial for VD and how it correlates to spatial scale events in scene. Another method [11], used weber local descriptor (WLD) and extended

it into improved WLD (IWLD) by adding the temporal components. They proposed sparse-based model for classification to control the coefficients reconstruction error and minimize it. They learned the class labels from the corresponding dictionary atoms based on sparse model. Furthermore, the detection of abnormal activities in the real-world surveillance also remained challenging. Cosar *et al.* [12] employed an integrated pipeline to incorporate the output generated from the object trajectory and pixel-based analysis. This information enabled the detection of abnormal behavior. These trajectories contained the speed and direction of object as well as its finer motion. However, the practice made on these features became tedious due to hand carried engineering.

B. VD Practice Using Deep Learning Methods

Recent works carried out in deep learning have achieved a tremendous attraction to solve several computer tasks and are also in practice for VD. For instance, Serrano *et al.* [8] developed a hybrid approach of two-dimensional (2-D) CNN and HF where the important information through HF are used to represent an image in a sequence. The HF is fed by the handcrafted features to capture the motion and appearance details for final VD. Similarly, a method presented in [13] used 3-D CNN to detect the violent scenes in videos where the sampling is applied as preprocessing and the key frames are used as dividing nodes. Next, Fenil *et al.* [14] introduced a spark framework where frames are processed and learned through bidirectional LSTM (Bi-LSTM) to detect the violent scenes in the football ground while Khan *et al.* [15] proposed a deep learning-based model to detect the violent scenes in videos by applying transfer learning approach to Mobile-Net and classified the violent and nonviolent scenes. Similarly, Akti *et al.* [16] utilized different LSTM-based approaches for fight detection in surveillance videos. Their approaches are based on LSTM, Bi-LSTM, and some other deep learning models including Xception and VGG16 to evaluate the detection performance. Most recently, Sing *et al.* [17] introduced the concept of aggregating the ensembles to detect the abnormalities in the crowded video scenes. Using different CNN architectures, their method learned several levels of semantic representation. They extracted the features using ConvNets and trained few variants of support vector machine (SVM) classifier. Another method [18] combined 3-D CNN with SVM to detect violent scenes in videos. Through deep insight of the literature, majority of the methods covered in the literature have several downsides and challenges, such as exclusion of IoT, usage of end-to-end pre-trained models, missing cloud-based concept, and practice of hand-feature engineering. These challenges and issues have made them more complex and computationally expensive for VD in surveillance videos.

III. PROPOSED IIOT-BASED VD-NET FRAMEWORK

In this section, we explain the proposed VD-Net framework in detailed fashion by investigating each step for its function that is visually explained in Fig. 1. The proposed IIoT-based framework consists of five basics steps. The step zero shows the procedure of training the light-weight CNN model for object

Algorithm 1: Pseudocode for training VD-Net (Offline phase).

Input: $\mathcal{D} \leftarrow$ Dataset

Initialization:

1. CLSTM $\leftarrow$ load (ConvLSTM)
2. GRU $\leftarrow$ load (GRU)
3. $\mathcal{S}_{TR} \leftarrow$ load (Training set)
4. $\mathcal{S}_{eTR} \leftarrow$ load (Training sequence)

Steps:

1. **for** $i = 1$ to $|\mathcal{S}_{TR}|$ **do**
2. $\mathcal{S}_{eTR} \leftarrow$ Read ($\mathcal{S}_{TR}^i$)
3. $F_{C-m} \leftarrow$ CLSTM ($\mathcal{S}_{eTR}$)
4. Prediction $\leftarrow$ GRU (F_{C-m})
5. **end for**

Output: VD Trained Model $\mathcal{D}_{mt}$.

Algorithm 2: Pseudocode of the proposed method (Online phase).

Inputs: $i\mathcal{V}_s \leftarrow$ Industrial surveillance video

Initialization:

1. $OD_t \leftarrow$ load (Object detection trained model)
2. $\mathcal{D}_{mt} \leftarrow$ load (VD trained model)
3. $\mathcal{S}_e \leftarrow$ load (Input sequence)

Steps:

1. **while** ($i\mathcal{V}_s$)
2. $\mathcal{S}_e \leftarrow$ read ($i\mathcal{V}_s$)
3. $F_{(obj)} \leftarrow OD_t(\mathcal{S}_e)$
4. **if** $F_{(obj)} = F_{(G,K)}$
5. Alert/share-details (IIoT Network) &&
 $\mathcal{D}_{mt}(F_{(G,K)})$
6. **else**
7. Repeat step 2
8. **end if**
9. **end while**

Output: Violent, Nonviolent

detection in offline fashion. The first step performs the job of data acquisition using resource constrained vision sensors while the second step contains the screening process for the important information collection, such as humans or suspicious objects in the scene. If the objects after detection in the second step, are humans or suspicious object, an alert is generated in the case of suspicious object by third step and the concerned frames are sent to the fourth step to deeply investigate the frames for final VD.

Algorithm 1 explains the detailed pseudocode of the proposed VD-Net, where the input dataset $\mathcal{D}$ is divided into training set $\mathcal{S}_{TR}$, testing set $\mathcal{S}_{TS}$, and validation set $\mathcal{S}_{VL}$. A sequence $\mathcal{S}_{eTR}$ from $\mathcal{S}_{TR}$ is fed into ConvLSTM that gives the features map F_{C-m} which is forward propagated into GRU for final prediction and after validation, the trained model $\mathcal{D}_{mt}$ is obtained.

Algorithm 2 explains the detailed pseudocode of the proposed VD-Net, where the input dataset $\mathcal{D}$ is divided into training set $\mathcal{S}_{TR}$, testing set $\mathcal{S}_{TS}$, and validation set $\mathcal{S}_{VL}$. A sequence

TABLE I
DATASET DETAILS FOR SUSPICIOUS OBJECT DETECTION

Dataset	Total images		Data splitting	
	Guns	Knives	Training set	Testing set
Guns and Knives [20]	300	300	480	120

[21]. Their captivating results have motivated the researchers for their usage in different problems. For this purpose, we investigate light-weight CNN in IIoT network to perform the screening process. The Raspberry Pi device screens every sixth frame in the video acquired through vision sensor for important information collection, such as humans or suspicious objects in industrial setup. If any suspicious object is detected, this information is shared with all the IIoT devices. Performing the practice of CNN-based algorithms for its precise and accurate performance over the resource constrained device is a difficult task due to their less computational power. Therefore, it is a big challenge to hold this process over minute edge device. To cope this problem over the edge device, we explore a light-weight object detection model by fine-tuning it in offline manner for the detection of suspicious objects and humans.

The suspicious objects include Guns and Knives [20] that consists of various types with different backgrounds. We fine-tune the latest Yolov3 tiny version with the pretrained ImageNet weights (Darknet53) using transfer learning approach for object detection that screens the frames for humans and suspicious objects. This tiny version efficiently processes the input frames on the edge. During training, the objects' area in the images of newly added classes are annotated. Next, the annotation details are generated according to the Yolov3 network settings and are saved in a separate text file that contains both the labels and dimensional details of the corresponding objects. The total number of images in the dataset are 600 for knives and guns that are divided into train set (80%) and test set (20%) where Table I gives the statistics of the dataset. After this setting, the input images, configured file, and model weights are given to the network. The model is trained with 4000 epochs with defined batch of 16. The network is modified in terms of the number of newly added N-classes and the filters size where the number of filters is reduced from 255 to 24, i.e., $(N + 5) \times 3$ where the number of newly objects are $N = 3$. We visually illustrated two testing samples in Fig. 2 for (a) knife and (b) gun with their detection results. Furthermore, the time analysis is covered in the experimental section.

C. Technical Details of VD-Net

In this section, we discuss and investigate the proposed VD-Net with its internal settings in detail fashion.

1) Encoding Mechanism: To identify and recognize a violent scene in a video, it is essential for a deep network to know the location of humans inside the frames and understand how their motion is changing over time. In such cases, CNNs are capable to generate good representation of each frame in the video. But, to encode the temporal changes in the frames, CNNs become inapplicable and require sequential learning technique to deal long-term temporal changes. Hence, recurrent neural networks

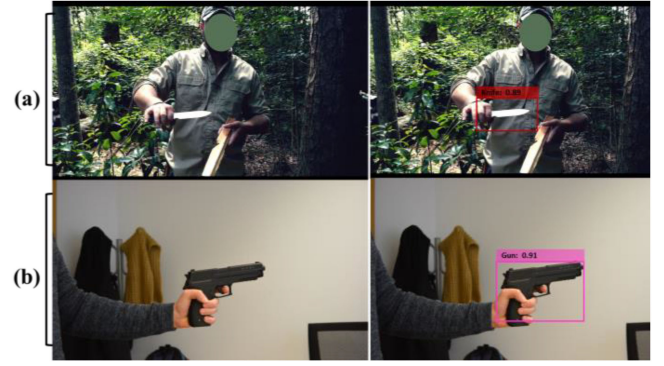


Fig. 2. Suspicious objects declared are guns and knives for training the light-weight CNN model. (a) Sample input and resultant image for knife class. (b) Sample input and resultant image for the gun class.

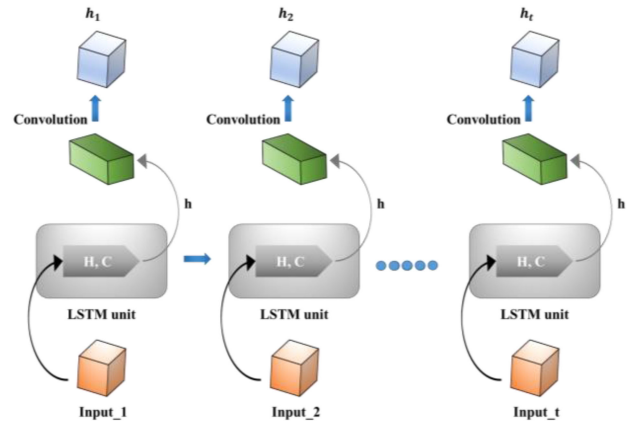


Fig. 3. General overview of a ConvLSTM cell. The cell states are passed into next LSTM cell in the flow direction. The black line represents the first input in certain time step.

(RNNs) are the popular sequential learning mechanisms to handle this problem. But we are interested in both spatial and temporal nature, to this purpose, ConvLSTM is the most suitable option. Next, existing techniques use fully connected (FC) layers (features) to encode data temporally via LSTM. The FC layer represents the global description of the frame, but it fails to encode the spatial changes which results to involve more data stream ensuring the computational complexity. Therefore, ConvLSTM is a better option that enables the network to learn the global and long-term spatiotemporal information of a video via convolutional gates (see Fig. 3) in them, without any shrinking in the size during intermediate representations. Similarly, the traditional (FC-LSTM) has the drawback of using full connections in transition of input to state and state to state where spatial information is not encoded. Therefore, the last two dimensions of all the inputs $P_1, P_2, \dots, P_n$, hidden states $h_1, h_2, \dots, h_n$, cell states $C_1, C_2, \dots, C_n$, and the gates i_t, f_t, o_t of ConvLSTM are spatial dimensions (rows/columns). ConvLSTM regulates future state of certain cell through inputs and previous states of local neighbors in the grid. Equations (1)– (5) show the formulation

held in ConvLSTM layers

$$i_t = \sigma(W_{pi} * P_t + W_{hi} * h_{t-1} + w_{ci} \circ C_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{pf} * P_t + w_{hf} * h_{t-1} + w_{cf} \circ C_{t-1} + b_f) \quad (2)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tanh(w_{pc} * P_t + w_{hc} * h_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo} * P_t + w_{ho} * h_{t-1} + w_{co} \circ C_t + b_o) \quad (4)$$

$$h_t = o_t \circ \tanh(C_t) \quad (5)$$

where “*” indicates the convolution operation while “ $\circ$ ” shows the Hadamard product. The hidden state is shown by “ h_t ”, “ c_t ” in the memory cell while “ i_t ”, “ f_t ”, and “ o_t ” are the activation gates. There are five layers of ConvLSTM with 3×3 filter size. After each convolutional LSTM, a dropout layer is added that regularize the output information to avoid overfitting problem.

2) Convolutional LSTM-GRU Network: The network to learns and identify the violent scenes is given in Fig. 1. The initial ConvLSTM layers are used to extract the hierarchical features from the video frames. Traditional SOTA techniques consider end-to-end pretrained model as a whole CNN-LSTM for learning purpose where the features are extracted and learnt by the network result into large number of parameters making the network complex. However, we are dealing with surveillance video patterns that are challenging where the indoor scenes, sometimes face the lightening conditions, therefore, its learning via simple network becomes difficult. Thus, assisting the ConvLSTM network, we plug an additional layer of GRU. GRU is similar to LSTM, but is simpler and easy to implement which reduces the gating signal into two from LSTM. The architecture of GRU is like LSTM except distinct ways to prevent vanishing gradient problem. The two gates in GRU include the update and reset gate. The reset gate functions are similar to forget gate of LSTM. The architecture of GRU is like LSTM except distinct ways to prevent vanishing gradient problem. However, GRU controls the information without using memory unit by exposing the hidden content. Next, it is relatively new and computationally more efficient as two gates perform the job of the process compared to LSTM [24]. Hence, to minimize the training error via avoiding the local minimal point, the Adam optimizer is used for back propagation through time. The weights corresponding to the gates are updated via backpropagation through time.

IV. DETAILED EXPERIMENTAL ANALYSIS AND DISCUSSION

In this section, we briefly discussed the system configuration, experimental results, and comparative analysis of our proposed VD-Net with SOTA. The empirical assessment of the proposed VD-Net is done using receiver operating characteristic (ROC), area under the curve (AUC), precision/recall, and confusion matrix. The details are given as follows.

A. System Configuration

The proposed framework is consisting of Raspberry Pi device as edge server to acquire data stream connected to IIoT network. The edge server used for screening is model B (version 3) with A53 having 1.2 GHz processor. The proposed VD-Net is formed

and implemented in Python (version 3.5) using Keras (version 2.3.1), a famous deep learning framework with Tensorflow (1.3.1) on backend and Adam as optimizer. Similarly, the cloud platform consists of RTX GPU GeForce 2080-Ti. To perform the experiments, the defined batch size is 8 and the model is trained with 150 epochs.

B. Datasets

Recently, the technology grabbed the strength and several sectors are engaged to proceed their work. However, it became difficult to monitor several activities for safety and security concerns through manual surveillance. To assist the system intelligently, we consider four datasets for VD model, i.e., surveillance fight dataset [16], RWF-200 [22], hockey fight dataset [23], and industrial surveillance dataset. Each dataset comprises of two classes, i.e., violent and nonviolent as shown in Fig. 4. Next, for experiments, we divided the datasets into 60% as training set, 20% as validation set, and 20% as testing set that is a standard splitting procedure. Moreover, the statistical and basic information of these datasets are discussed in the following sections.

1) Existing Benchmarks: The surveillance fight dataset [16] contains indoor, outdoor, night, and day-time videos, gathered from real-world surveillance and different sources (YouTube, etc.). Similarly, RWF-2000 Dataset [22] also contains videos taken in indoor, outdoor, day, and night scenarios in factories, working places, etc. This dataset is totally consisting of surveillance videos and none of the videos are modified through multimedia technology. Furthermore, the hockey fight dataset [23] is collected over the hockey grounds (National Hockey League). We visually shown few samples of each dataset in Fig. 4 while their statistical details are given in Table II.

2) New Industrial Surveillance Dataset: We collected this dataset from different sources and search engines such as YouTube and Google by inserting diverse queries, such as violence scenes in industrial surveillance, in factories, and in steel mills. The obtained videos from different sources have distinct video resolution and frame rate. The length of retrieved videos ranges from 7 to 12 min that are trimmed to 5 s violent and nonviolent clips for each class. We arranged the dataset in the same standard format, such as surveillance fight dataset. The industrial dataset consists of varied scenes recorded in different industries, stores, offices, and petrol pumps linked to industries. Compared to existing datasets, the industrial surveillance dataset is more challenging because most of the actions are aside of the center point from the camera and the fps varies like other surveillance datasets. Its statistical details and sample frames are given in Table II and Fig. 4(d), respectively.

C. Results and Discussion

This section discusses the detailed evaluation of the proposed VD-Net via the experiments performed on each dataset. Next, to fairly compare the superiority of the proposed VD-Net over the conventional ConvLSTM, the ConvLSTM is also trained and evaluated on each dataset. These details are as follows.



Fig. 4. Visual representation of the samples taken from both classes of each VD dataset. (a) Sample frames obtained from surveillance fight dataset where first three columns are related to the violent while the nonviolent frames are given in next three columns. (b) Violent and nonviolent frames from hockey fight dataset (c) Sample frames from RWF-2000 dataset that consist of the real-world surveillance videos. (d) Sample frames from both classes of industrial surveillance dataset.

TABLE II
STATISTICAL OVERVIEW OF VD DATASETS UTILIZED FOR THE EVALUATION OF THE PROPOSED VD-NET

Dataset	Total samples	Frame resolution	Clip length (Seconds)	Violent		Nonviolent	
				Samples	Frame rate	Samples	Frame rate
Surveillance Fight [16]	300	Variable	2	150	20–30	150	20–30
RWF-2000 [22]	2000	Variable	5	1000	30	1000	30
Hockey Fight [23]	1000	360 × 240	1.6–1.96	500	25	500	25
Industrial Surveillance (Ours)	300	Variable	5	150	20–30	150	20–30

TABLE III
DETAILED EVALUATION RESULTS OF THE CONV LSTM AND PROPOSED VD-NET BASED ON ACCURACY, AUC, PRECISION, RECALL, AND F1-SCORE USING PUBLIC BENCHMARKS WHERE THE BOLD VALUES REPRESENT THE GOOD PERFORMANCE

Method	Dataset	Obtained values				Accuracy (%)	AUC	Precision	Recall	F1-score
		TP	TN	FP	FN					
ConvLSTM	Surveillance Fight [16]	18	18	13	9	62	0.584	0.581	0.667	0.621
	RWF-2000 [22]	158	183	42	17	85.3	0.867	0.813	0.915	0.861
	Hockey Fight [23]	92	96	6	6	94	0.982	0.941	0.941	0.941
	Industrial Surveillance (Ours)	23	21	8	8	73	0.824	0.724	0.724	0.724
Proposed VD-Net	Surveillance Fight [16]	23	21	8	6	75.9	0.821	0.724	0.778	0.750
	RWF-2000 [22]	167	186	33	14	88.2	0.91	0.849	0.930	0.888
	Hockey Fight [23]	97	100	1	2	98.5	0.994	0.990	0.980	0.985
	Industrial Surveillance (Ours)	24	24	7	5	80	0.846	0.774	0.828	0.800

1) *Evaluation Using ConvLSTM*: The experiments performed using ConvLSTM are given in Table III. First, we practice each dataset for its training using ConvLSTM network to verify and examine the network performance. Existing methods have widely used end-to-end networks for classification purposes. They have used trained CNN architectures and have performed learning using conventional RNN techniques. Due to this purpose, these methods are computationally complex and produce large number of parameters. However, the proposed VD-Net has layers of ConvLSTM. To deeply investigate it, we consider ConvLSTM layered approach and perform ablation study to compare its results with the proposed VD-Net. The accuracy obtained for ConvLSTM on surveillance fight,

RWF-2000, hockey fight, and industrial surveillance dataset is 62%, 85.3%, 94%, and 73%, respectively. To hold a deep investigation of correctly and wrongly classified classes, we measure the confusion matrix values, such as true positive (TP), true negative (TN), false positive (FP), and false negative (FN) that also give Precision, recall, and F1 score which are important metrics. We also construct the ROC for ConvLSTM and compare its performance with proposed VD-Net that shows its superiority over ConvLSTM. This comparison in terms of AUC and ROC for surveillance fight, RWF-2000, hockey fight, and industrial surveillance dataset is visually presented in Fig. 5(a)–(d) where the tradeoff between the TP-rate and FP-rate is shown through ROC. This curve is drawn via TP-rate against FP-rate with X-axis

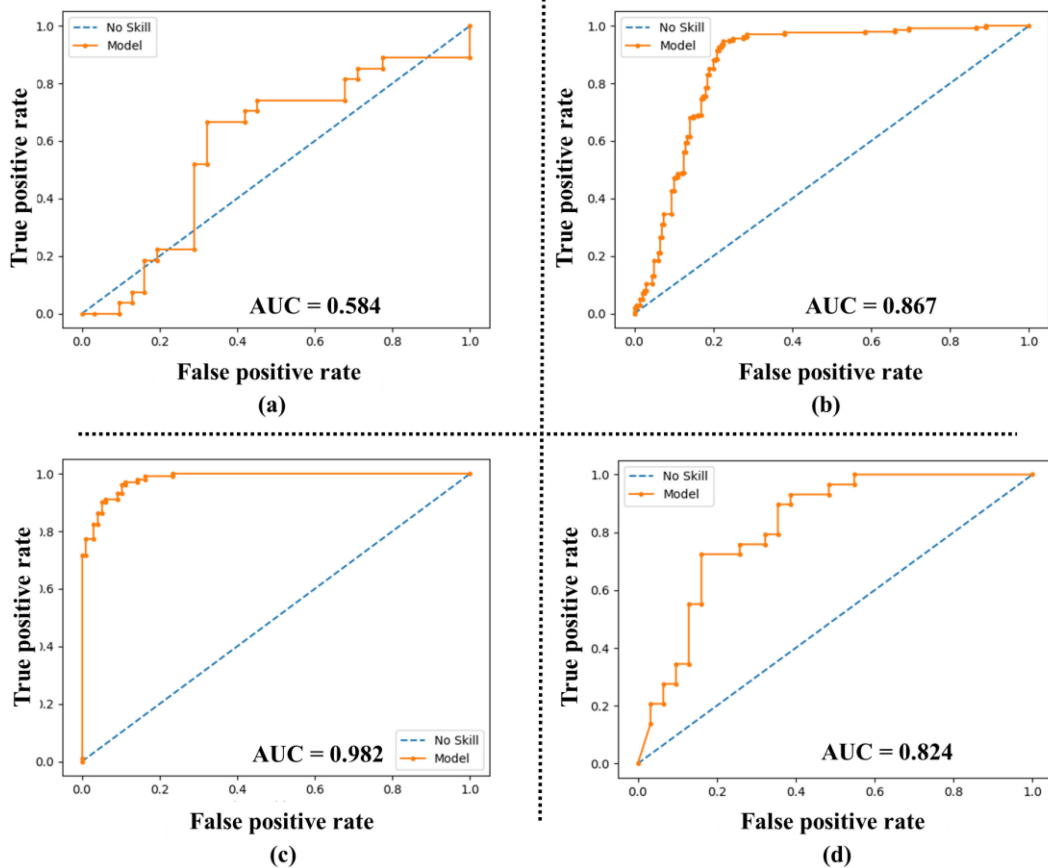


Fig. 5. Detailed analysis of the ConvLSTM using AUC and ROC. (a) ROC and AUC for surveillance fight dataset. (b) ROC and AUC obtained for RWF-200. (c) Results for hockey fight. (d) Performance of industrial surveillance dataset.

and Y-axis, respectively. Similarly, the AUC is also calculated which states that an ideal model with value near to 1 shows better performance.

2) Evaluation Using the Proposed VD-Net: As the technology grow with time passage, the protection and safety of humans along with their property assets over industrial setup faces their security issues. To combat this, the smart city and the modern industrialized world have installed the surveillance systems to enable the layer of activity monitoring. Automating this system, different issues arises as most of the existing methods are based on nonsurveillance-based domain, i.e., usage of nonsurveillance datasets. Therefore, aiding this technology, we broadly focus on VD-based surveillance setup and evaluate our method on recently published datasets [16], [22] to compare its performance with SOTA. We created our own dataset known as industrial surveillance dataset which plays a benchmarking role for secure industrial surveillance. Next, we also compare the results for hockey fight [23] (nonsurveillance) with existing methods due to its challenging nature. For this purpose, this section deeply discusses the evaluation of the proposed VD-Net on each dataset and investigates SOTA for deep learning techniques. Similar to ConvLSTM, we evaluate the proposed VD-Net using four datasets and their results are given in Table III and the confusion matrix is shown in Fig. 6. The accuracy obtained for Surveillance Fight, RWF-200, hockey fight, and industrial surveillance

dataset is 75.9%, 88.2%, 98.5%, and 80%, respectively. Next, the AUC and ROC for surveillance fight, RWF-2000, hockey fight, and industrial surveillance dataset is visually presented in Fig. 7(a)–(d), respectively. Sometimes, the VD methods considers any progressive action as a violent, this definition arises typically in sport as an issue. For instance, in sports, such as hockey fight videos, the colliding of players happens or hitting each other. So one strategy is to detect violence is to identify when the player moves closer to another. However, nonviolent scenes also comprise of players where they are hugging for victory celebration, etc., that has high chance of mistaking nonviolent as violent. Nevertheless, the proposed VD-Net is capable to avoid this mistake by encoding the localized regions of motions, such as motion of hands and limbs.

3) Time Complexity Analysis: 1) *Cloud Server:* The processing time for VD in IIoT assisted surveillance is very important to consider. In this case, the proposed VD-Net obtained promising performance to process the sequence. The ConvLSTM and VD-Net process a sequence of 28 and 62 frames per second, respectively. Hence, to process one frame, the proposed VD-Net takes 0.01612 s that is 2.21 times faster than ConvLSTM which confirms and guarantees the effectiveness of proposed VD-Net. Similarly, the number of obtained parameters on ConvLSTM and VD-Net are 18 976 770 and 4 470 298, respectively, that shows a great achievement of the proposed VD-Net.

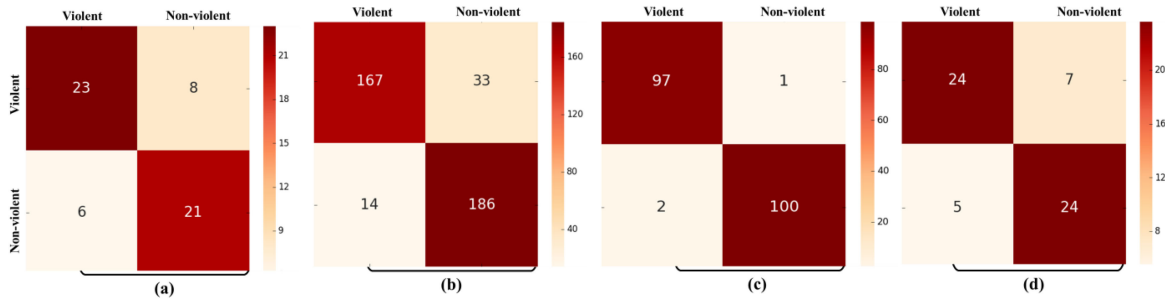


Fig. 6. Visual representation of the confusion matrix for the proposed VD-Net over each dataset. (a) Visual representation of the confusion matrix for surveillance fight. (b) Confusion matrix obtained for RWF-2000. (c) Confusion matrix for hockey fight. (d) Performance results of industrial surveillance dataset.

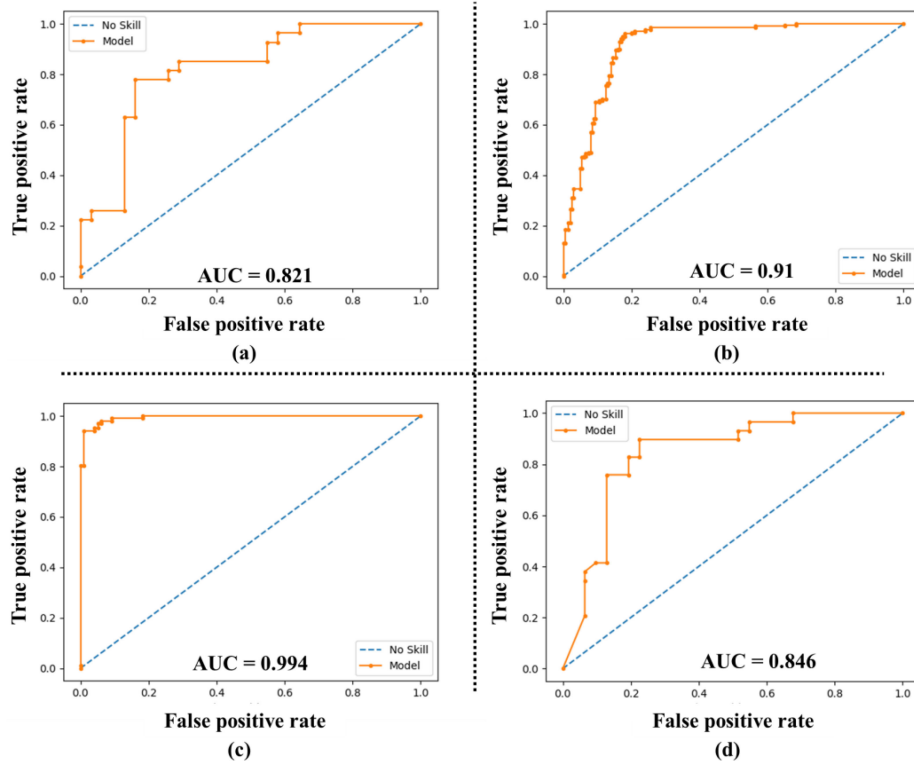


Fig. 7. Detailed analysis of the proposed VD-Net using AUC and ROC. (a) ROC and AUC for surveillance fight dataset. (b) ROC and AUC obtained for RWF-200. (c) Results for hockey fight. (d) Performance of industrial surveillance dataset.

2) *Edge Server*: In the proposed IIoT-based VD-Net framework, the edge server (Raspberry Pi) is responsible for the suspicious object detection along with Movidius NCS. The model on this server takes 1 s to process six frames. However, we process every sixth frame from the live video stream. Thus, the total time taken by the CNN model to process one frame on the edge is 0.1666 s while its fps on cloud is 250. Similarly, the mean average precision obtained is 46.26.

D. Comparative Analysis of the Proposed VD-Net

This section extensively carries out the comparative study of the proposed VD-Net to verify and endorse its effectiveness

over the SOTA VD methods. Next, by thorough investigation of each method, we categorize them for their learning strategy. The detailed summarized representation of comparisons and the categorization of the learning strategy of the methods are given in Table IV. First, a method developed in [25] is discussed where they used violent flow (VF) descriptor and classified the violent and nonviolent scenes using SVM and obtained 82.90% accuracy on hockey fight dataset. Another method [11] proposed IWLD to depict low level appearance of image information. They extended the descriptor IWLD through adding temporal component for local motion information forming motion-IWLD and obtained 96.8% accuracy on hockey fight dataset. Similarly, another method [26] presented elastic cubic trajectories to detect

TABLE IV

COMPARATIVE ANALYSIS OF THE PROPOSED VD-NET WITH SOTA DATASETS USING SURVEILLANCE FIGHT, RWF-200, AND HOCKEY FIGHT, DATASET IN TERMS OF ACCURACY (%) AND THEIR LEARNING STRATEGIES. EACH METHOD IS ASSIGNED BY SPECIFIC ID WHERE PM SHOWS THE PROPOSED METHOD

ID	Ref.	Method	Learning strategy		Surveillance fight (%)	RWF-2000 (%)	Hockey fight (%)
			Deep learning	Conventional learning			
M1	[25]	VF-SVM	×	×	×	×	82.90
M2	[11]	IWLD	×	✓	×	×	96.8
M3	[26]	Cuboid Trajectories + HF	×	✓	×	×	82.5
M4	[8]	HF + 2D CNN	✓	✓	×	×	94.6
M5	[27]	OVIF + HOMO + SVM	×	✓	×	×	89.3
M6	[15]	Mobile-Net	✓	×	×	×	87.2
M7	[28]	MoBSIFT	×	✓	×	×	96.5
M8	[29]	HOG3D + BoVW	×	✓	×	×	95
M9	[30]	MMPU	×	✓	×	×	87.3
M10	[22]	Flow Gated Network	✓	×	×	87.25	98
M11	[16]	FightCNN + BD-LSTM	✓	×	72	×	98
PM		VD-Net (proposed)	✓	×	75.9	88.2	98.5

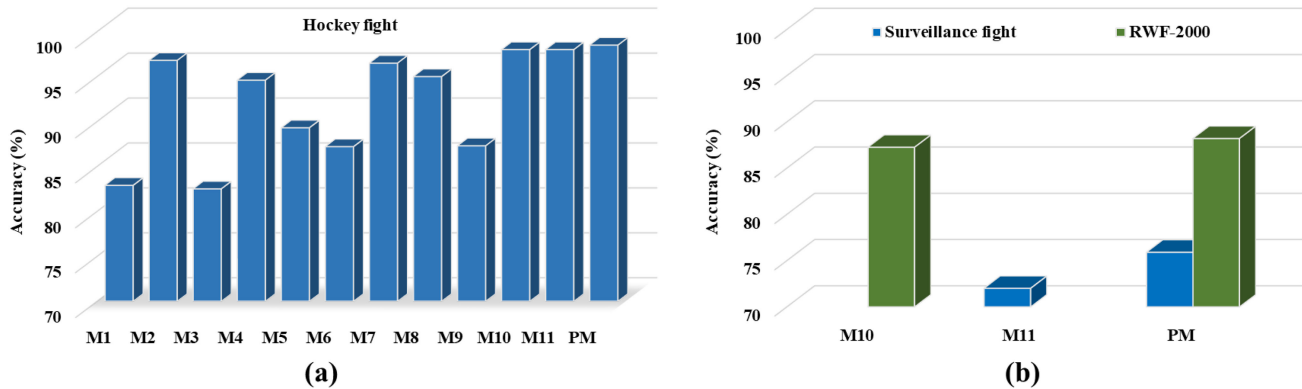


Fig. 8. Comparative analysis of the proposed method with SOTA methods. The methods are assigned with specific ID and their corresponding results are given in Table IV. (a) Hockey fight results and its comparison with SOTA methods. (b) Both the surveillance datasets are given with their results showing superiority of the proposed method over existing methods.

fight by creating the trajectories through blobs movement which capture different motions specific to the fight. They adapted HF classifier with specific descriptor and achieved 82.5% accuracy on hockey fight dataset. Furthermore, a hybrid method comprises of HF and 2-D CNN [8] summarized the video contents into resultant image that encoded the appearance and leveraged motion areas which represent the fight or aggressive behavior. They used 2-D CNN that discriminates between the previous obtained images between aggressive and normal situation. They conducted different experiments by the combination of several features descriptor to boost the accuracy and obtained 94.6% as highest accuracy on hockey fight dataset. Next, a method in [27] introduced a descriptor histogram of optical flow magnitude and orientation (HOMO) and oriented violent flow (OVIF) for features collection from the input images and converted them into grayscale before computing the optical flow. The extracted information is learned through SVM to classify the violent and nonviolent class and achieved 89.3% accuracy. A VD method in [15] used Mobile-Net architecture to detect the violent scenes in movies and obtained 87.2% accuracy on hockey fight dataset. Similarly, in [28], scale invariant feature transform

(SIFT) with histogram optical flow feature and motion boundary are extracted and combined with the motion boundary SIFT (MoBSIFT) and achieved 96.5% accuracy. Another method [29] applied the approach that consists of three dimensional histograms of gradient orientation (HOG3D), Bag BoWs, and feature pooling technique. They extracted the HOG3D features on the block level and the BoW is trained where the features pooling technique is operated for the entire video segment and obtained 95% accuracy on the hockey fight dataset. A method presented in [30] developed a positive/unlabeled (PU) learning mechanism for visual analysis known as multimanifold PU (MMPU). They applied different vision related tasks where the detection of violent behavior is also considered to evaluate their method. Using this learning mechanism, 87.3% was the highest obtained accuracy on hockey fight. The overall visual evaluation on hockey fight dataset is graphically presented in Fig. 8(a).

The comparison is also performed with surveillance-based methods that use RWF-2000 [22] and surveillance fight [16]. The method in [22] used flow gated network by applying ablation study and introduced RWF-2000 dataset. They obtained

87.25% and 98% accuracy on RWF-2000 and hockey fight dataset, respectively. Similarly, the method in [16] is based on surveillance setup that used and explored different LSTM-based techniques. They integrated the Xception model, BD-LSTM, and attention model to check the performance and achieved 72% accuracy on surveillance fight dataset. Finally, we pose the results of proposed VD-Net where the highest accuracy achieved for surveillance fight, RWF-2000, and hockey fight datasets is 75.9%, 88.2%, and 98.5%, respectively. The overall visual evaluation on surveillance fight and RWF-2000 dataset is graphically presented in Fig. 8(b).

V. CONCLUSION

Over the decades, the improvement in the technology and surveillance infrastructure was observed for security and assets' protection in the public and industrial settings. Monitoring such places through manual ways was a tedious and time-consuming process for activity analysis and violent scenes detection. Toward this problem, existing methods were applying distinct algorithms for its solution and process the overall video frames at the local server and exclude IIoT settings. To tackle this challenge, in this article, we introduced an IIoT-based framework by proposing VD-Net that was capable of detecting the violent scenes over industrial setup using AI techniques. The input video frames acquired through vision sensors were passed into light-weight CNN to search for important information, such as humans and suspicious objects. In case of suspicious object detection, the information was shared in the IIoT network. Next, only the important frames were sent into cloud server where features using ConvLSTM were extracted and fed into GRU for final VD. We significantly improved the accuracy over the existing literature with the margin of 3.9%, 0.95%, and 0.5% for surveillance fight, RWF-2000, and hockey fight dataset, respectively. Similarly, the performance obtained over the new benchmark industrial surveillance dataset confirmed our method's suitability in industrial setup and showed that the proposed VD-Net was more effective for its deployment in industrial surveillance systems.

In future work, we aim to strengthen the VD process by considering the multiview data to evaluate it for detailed investigation because the existing methods are widely based on single view where the single camera is unable to cover the whole scene. Therefore, in terms of multiview, the activity from every angle can be monitored thoroughly. Furthermore, nowadays, the technology has been growing to automate the recognition of different activities through fusion of multistream networks. Therefore, we aim to investigate fusion of different networks for VD in surveillance systems.

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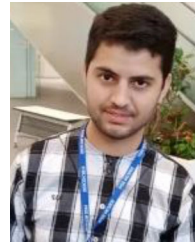
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